GNN for Fraudulent Pattern Detection

How do Graph Neural Networks identify money-laundering patterns in a transaction network?

Progress meeting



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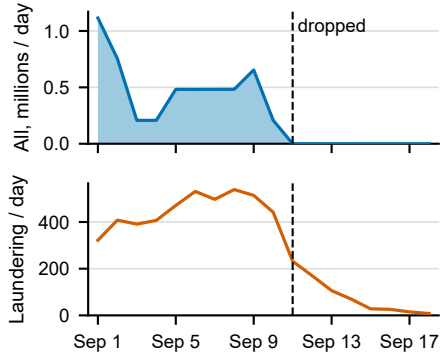
References

Problem and goal

- **Task:** classify each transaction in a transaction graph as laundering or not
- **Question:** can GNNs find laundering patterns, and do graph features help?

Dataset

Property	Value
Source	HI-Small (Altman et al., 2023)
Transactions	5,078,345 → 5,077,237
Laundering	4,522 (0.089 %)
Accounts (nodes)	515,070
Time span	2022-09-01 → 2022-09-10
Currencies	15, converted to USD
Annotated attempts	370, in 8 typologies
Split by time	60 % train, 20 % val, 20 % test

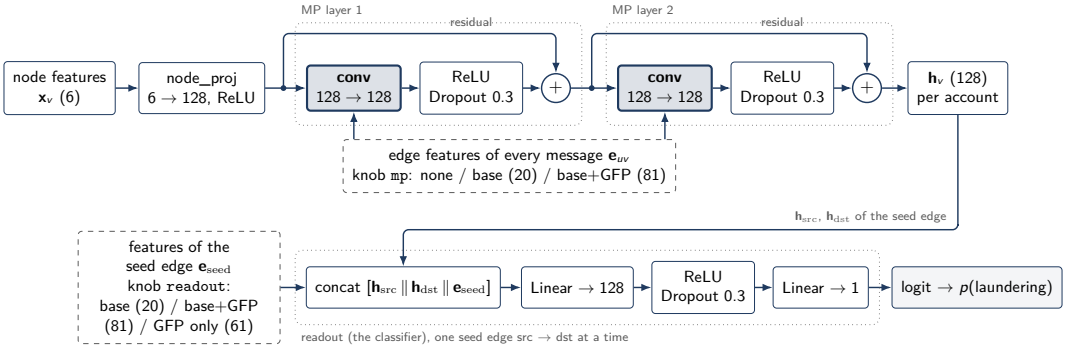


- **Real data ends Sep 10; later days hold only launderers' transactions**

Features

Feature	Dim	What it captures
Node features (6 per account)		
`EntityType_*`	6	kind of account holder (one-hot)
Baseline edge features (20 per transaction)		
`Amount_Log`	1	amount in USD, log scale
`Struct_Band`	1	amount just below the $10k reporting threshold
`Hour_Sin/Cos`, `DayOfWeek_Sin/Cos`	4	time of day and day of week (cyclic)
`Is_Weekend`	1	weekend flag
`Is_Self_Loop`	1	sender and receiver are the same account
`Same_Bank`	1	sender and receiver at the same bank
`Dt_Src_Log`, `Dt_Dst_Log`	2	time since the sender's / receiver's previous transaction
`Src_Bank_Risk`, `Dst_Bank_Risk`	2	past laundering rate of the sender's / receiver's bank
`PayFmt_*`	7	payment channel, one-hot (ACH, Bitcoin, Cash, Cheque, Card, Reinvestment, Wire)
GFP edge features (61 per transaction)		
Scatter–gather	3	scatter–gather patterns the transaction belongs to, by size
Temporal cycles	3	time-ordered cycles the transaction closes, by length
Simple cycles	3	cycles up to length 6 the transaction closes, by length
Vertex statistics	52	fan, degree, amount and timing statistics of sender and receiver

Fixed architecture



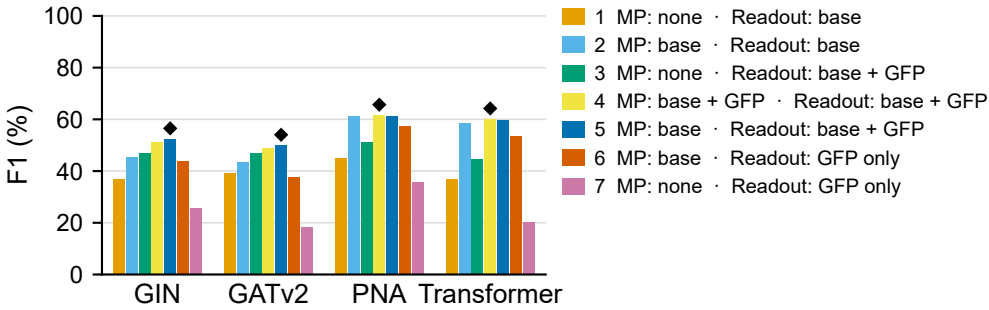
- **Only the shaded conv layer changes between runs**

Seven feature configurations and training setup

Configuration	1	2	3	4	5	6	7
Edge features in MP	none	base	none	base+GFP	base	base	none
Edge features at readout	base	base	base+GFP	base+GFP	base+GFP	GFP only	GFP only

Training setting	Value
Neighbour sampling	100 per hop, 2 hops
Batch	8,192 seed edges
Loss	BCE, pos_weight 8
Optimiser	Adam
Learning rate	10^{-3}, cosine schedule
Epochs	20
Hidden width	128
Message-passing layers	2
Dropout	0.3
Checkpoint	best validation F1
Seed	42

All configurations per operator



MP = edge features inside message passing; Readout = edge features given to the final classifier;
base = 20 transaction features; GFP = 61 graph-pattern features; diamond = highest test F1 of the operator.

- **Edge features in message passing help every operator**

Test F1 and PR-AUC of all runs

Configuration	1	2	3	4	5	6	7
MP	none	base	none	base+GFP	base	base	none
Readout	base	base	base+GFP	base+GFP	base+GFP	GFP only	GFP only
F1 (%)							
GIN	37.1	45.6	47.1	51.2	52.5	43.9	25.6
GATv2	39.2	43.5	46.9	48.8	50.0	37.9	18.4
PNA	45.2	61.4	51.2	61.7	61.1	57.5	35.6
Transformer	37.0	58.6	44.8	60.2	59.8	53.4	20.4
PR-AUC (%)							
GIN	32.8	39.2	43.5	47.9	48.4	39.5	19.8
GATv2	33.0	38.9	41.9	45.8	45.1	31.2	11.2
PNA	38.9	57.8	47.2	57.5	57.4	54.0	29.0
Transformer	33.7	54.9	41.1	57.4	55.3	50.8	13.9

- **PNA and the Transformer lead on both F1 and PR-AUC**

How our runs compare with the dataset paper (HI-Small)

	Edge features		This work	Altman et al. 2023*
Model	MP	Readout	*F1 (%), one seed*	*F1 (%), 4 seeds*
GIN	base	base+GFP	52.5	28.7 ± 1.1
GATv2	base	base+GFP	50.0	NA
PNA	base	base	61.4	56.8 ± 2.4
Transformer	base+GFP	base+GFP	60.2	NA
GFP + LightGBM	–	base+GFP	NA	62.9 ± 0.3
GFP + XGBoost	–	base+GFP	NA	63.2 ± 0.2

* The paper's GNNs use 4 base edge features (timestamp, amount received, received currency, payment format) in both MP and readout, and a constant node feature of 1s.

- **We beat the paper's GIN and PNA, but not its tree models**

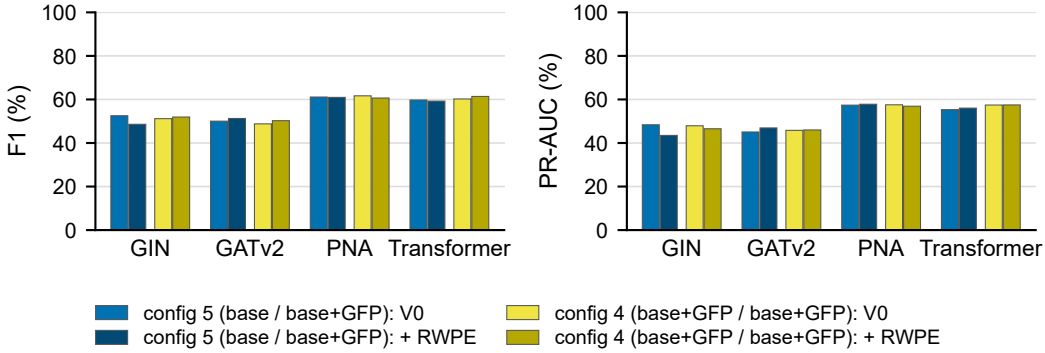
RWPE node encodings: what and why

RWPE	train	val	test
Accounts on a cycle	0.58 %	1.02 %	1.63 %

Cycle length to cover	hops
GFP cycle cap (V0)	6
GFP cycle cap (paper)	10
Longest annotated cycle	12
RWPE walk length k	16

- **The random walk follows the direction of the money**
- **Walk length 16 covers the longest cycle, 12 hops**

RWPE results: with vs without



- **No clear uplift from RWPE on test F1 or PR-AUC**

GFP V1, tuned to the data: settings

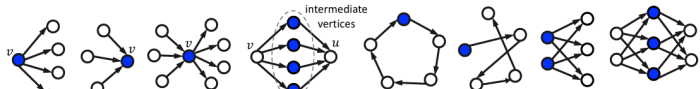
GFP feature group	What it counts	#	V0	V1
Scatter–gather	patterns completed, by size	3 / 4	6 h	12 h
Temporal cycles	time-ordered cycles closed, by length	3 / 4	24 h	48 h
Simple cycles	cycles closed, any order, by length	3 / 4	24 h, $\leq$ 6 hops	48 h, $\leq$ 12 hops
Pattern bins	size / length bins of the three groups above	–	[2, 3, 5] $\rightarrow$ 3 bins	[2, 4, 6, 8] $\rightarrow$ 4 bins
Vertex statistics	fan, degree, ratio; avg, sum, var, skew, kurt of amounts and times (sender / receiver, in / out)	52 / 52	24 h	48 h
Graph memory	memory of past transactions	–	24 h	48 h
Total per transaction		61 / 64		

Why tune GFP: the 370 annotated attempts

Typology	median	$\leq$ 24 h *V0*	$\leq$ 48 h *V1*	$\leq$ 120 h
BIPARTITE	24 h	47 %	100 %	100 %
CYCLE	72 h	9 %	22 %	100 %
FAN-IN	85 h	15 %	22 %	100 %
FAN-OUT	77 h	21 %	23 %	100 %
GATHER-SCATTER	151 h	16 %	16 %	31 %
RANDOM	46 h	41 %	51 %	100 %
SCATTER-GATHER	89 h	2 %	11 %	100 %
STACK	73 h	21 %	33 %	100 %
ALL (370)	75 h	21 %	35 %	91 %

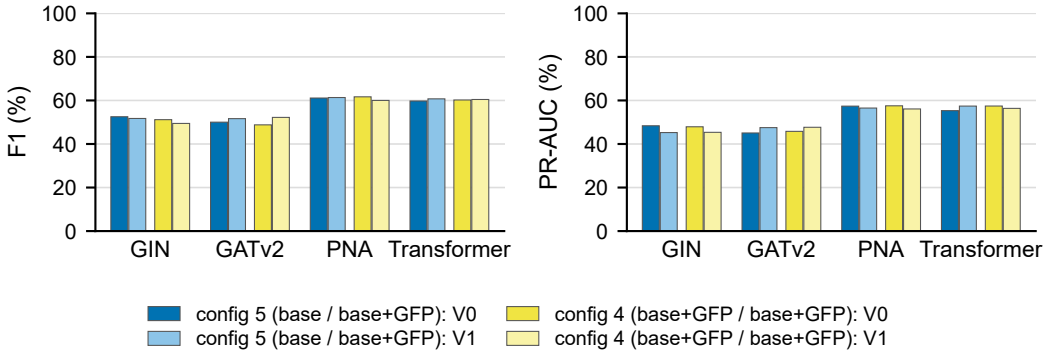
Cycle length (54)	
range	2–12 hops
longer than 6	37 %
Fan size (88)	
median	7.5
max	16

- **Seen whole: V0 21 %, V1 35 %**
- **Cycles reach 12 hops; V0 stops at 6**



(a) Fan-out (b) Fan-in (c) Gather-scatter (d) Scatter-gather (e) Simple cycle (f) Random (g) Bipartite (h) Stack

GFP results: V0 vs V1



- **Only GATv2 gains from V1; the other operators show no clear gain**

All 16 extension runs: test F1 and PR-AUC

		F1 (%)			PR-AUC (%)		
Operator	Config	V0	V0 +RWPE	V1	V0	V0 +RWPE	V1
GIN	5	52.5	48.6	51.7	48.4	43.5	45.2
GIN	4	51.2	51.9	49.4	47.9	46.5	45.4
GATv2	5	50.0	51.3	51.6	45.1	46.9	47.5
GATv2	4	48.8	50.2	52.2	45.8	46.0	47.7
PNA	5	61.1	61.0	61.3	57.4	57.8	56.5
PNA	4	61.7	60.7	60.0	57.5	56.9	56.1
Transformer	5	59.8	59.2	60.7	55.3	56.0	57.4
Transformer	4	60.2	61.4	60.4	57.4	57.5	56.4

- **Only GATv2 improves with both RWPE and V1**

Summary and next steps

Takeaways

- One fixed architecture makes all 44 runs comparable
- Edge features in message passing lift every operator, most of all PNA and the Transformer
- PNA and the Transformer lead; GFP helps only GIN and GATv2
- RWPE and GFP V1 improve only GATv2

Next steps

1. Investigate why GFP V1 does not lift GIN and GATv2 further
2. Per-typology evaluation of every operator and config
3. Repeated runs with different seeds for a robust estimate
4. Global attention: a Transformer over the whole graph, not just local neighbours
5. Node2Vec node embeddings
6. GFP V2 with longer time windows
7. Thesis writing

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